CHAPTER 09

# Carboxylic Acid Derivatives

## Content and rendering review packet

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

Source snapshot SHA-256: 879cc182a549d26acc8bc2878a40fac08c888a695dc824071e43aa4bbe7f4c4d

Review all claims, molecule graphs, lessons, exercises, hints and displayed states. Keys are author claims, not ground truth. Screenshots show static states with reduced motion: they do not prove animation timing or persistence behavior.

For every finding: item ID/page; severity (critical/major/minor); exact claim; derivation or reference evidence; proposed correction; confidence. List unreviewed/uncertain items. Model agreement is not scientific verification.

### Coverage checklist

ch09:exercise:cyclic-derivative | cyclic-derivative:stage-1, cyclic-derivative:stage-2, cyclic-derivative:stage-3

ch09:exercise:acyl-order | acyl-order:stage-1, acyl-order:stage-2, acyl-order:stage-3

ch09:exercise:hydrolysis-products | hydrolysis-products:stage-1, hydrolysis-products:stage-2, hydrolysis-products:stage-3

ch09:lesson:families | families:rule, families:cue, families:trap, families:app, families:step-1, families:step-2, families:step-3

ch09:lesson:acyl-mechanism | acyl-mechanism:rule, acyl-mechanism:cue, acyl-mechanism:trap, acyl-mechanism:app, acyl-mechanism:step-1, acyl-mechanism:step-2, acyl-mechanism:step-3

ch09:lesson:derivative-reactions | derivative-reactions:rule, derivative-reactions:cue, derivative-reactions:trap, derivative-reactions:app, derivative-reactions:step-1, derivative-reactions:step-2, derivative-reactions:step-3

ch09:molecule:ester | ester

ch09:molecule:betalactam | betalactam

ch09:molecule:lactone | lactone

ch09:molecule:methylacetate | methylacetate

ch09:molecule:ester-intermediate | ester-intermediate

ch09:molecule:anhyd | anhyd

ch09:molecule:acetamide | acetamide

ch09:curriculum | scope-1, scope-2, scope-3

# Curriculum scope

scope-1: 9.1 Amides, Esters, and Anhydrides

scope-2: 9.2 Reactivity Principles

scope-3: 9.3 Nucleophilic Acyl Substitution Reactions

Objective: Identify esters, amides, anhydrides, lactones, and lactams

Objective: Compare acyl substitution reactivity

Objective: Predict hydrolysis and transesterification outcomes

Supplied Kaplan curriculum: Chapter 9; section scope checked against local body and summary | organic_chemistry_kaplan/OEBPS/chapter09b.xhtml

No source passage is included. Request relevant authoritative excerpts when a claim cannot be checked by derivation.

# families | Recognize acid derivatives

Read the atom directly attached to acyl carbon

## families:rule | Core Rules

Esters have C(=O)–OR; amides have C(=O)–NR2; anhydrides have C(=O)–O–C(=O). Lactones are cyclic esters and lactams are cyclic amides. A beta-lactam contains a four-membered ring.

## families:cue | Memory Cues

Acyl C → O–C: ester. Acyl C → N: amide. Acyl C → O → acyl C: anhydride.

## families:trap | Common Mistakes / MCAT Traps

An ether has no adjacent carbonyl. An N-substituted amide is still an amide even if it has no N–H bond. Ring size names count heteroatoms as ring members.

## families:app | MCAT Applications

Triacylglycerols contain ester links; proteins contain amide links. Beta-lactam antibiotics exploit a strained amide structure.

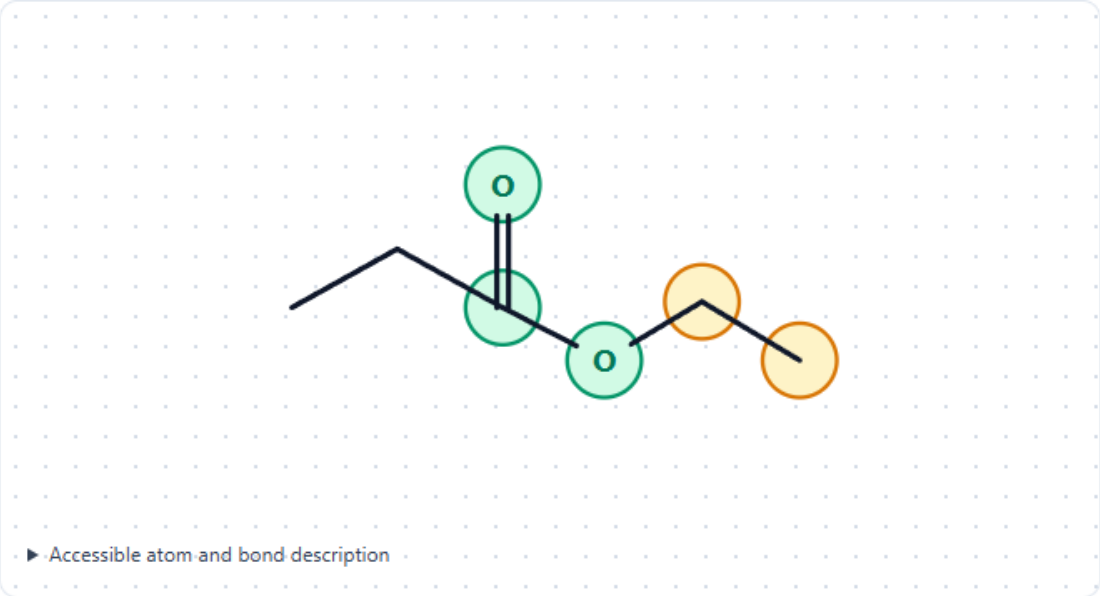
## families:step-1 | 1. Ester vs. amide

Ethyl propanoate connects the acyl carbon to oxygen. The alkoxy carbon chain lies beyond that oxygen.

Displayed molecule ID: ester

{"highlights": {"prin": [2, 3, 4], "sub": [5, 6]}, "revealName": true}

Screenshot: families-step-1



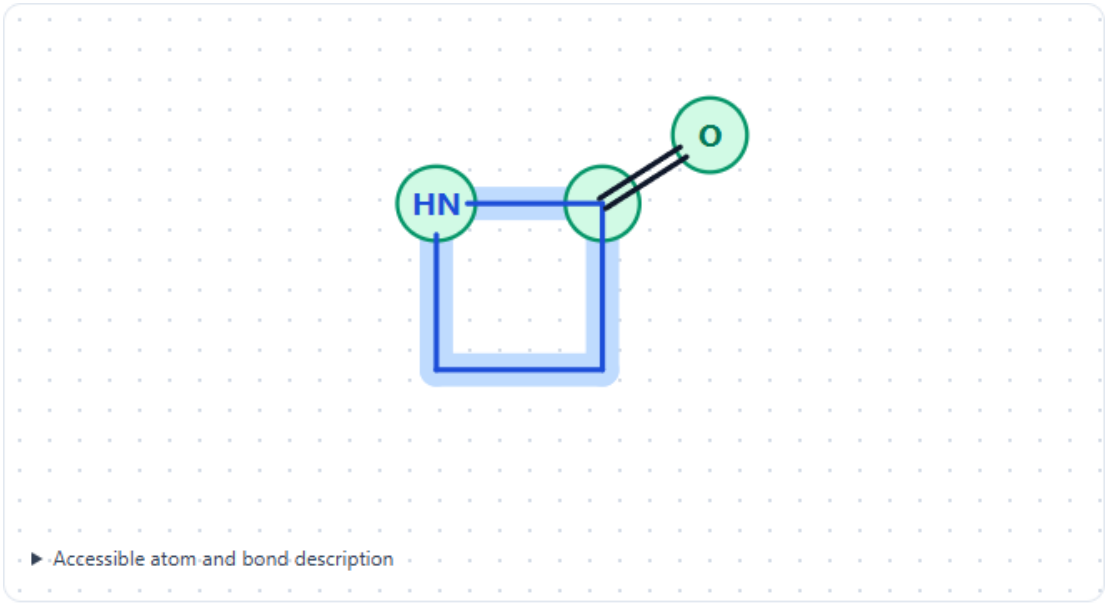

## families:step-2 | 2. Cyclic amide

The beta-lactam has four ring atoms, one of them N. The carbonyl carbon is part of the ring and the oxygen is outside it.

Displayed molecule ID: betalactam

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Screenshot: families-step-2



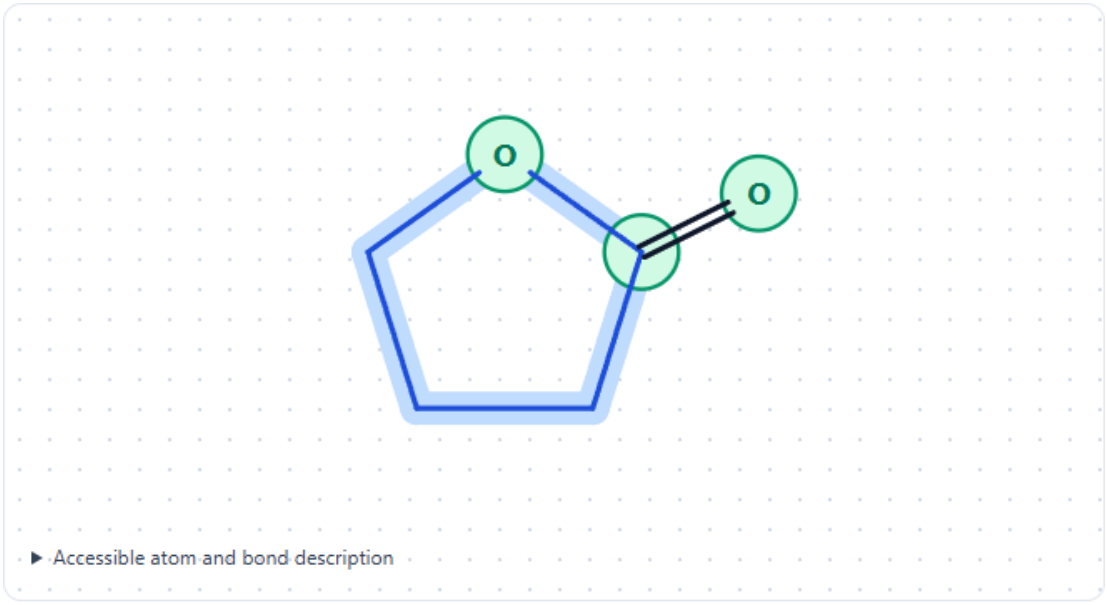

## families:step-3 | 3. Cyclic ester

A lactone contains an ester linkage within a ring. This five-membered lactone has one ring oxygen and four ring carbons.

Displayed molecule ID: lactone

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Screenshot: families-step-3



# acyl-mechanism | Acyl substitution and relative reactivity

Addition → tetrahedral intermediate → elimination

## acyl-mechanism:rule | Core Rules

A nucleophile adds to an acyl carbon and shifts the pi pair to O. Collapse of the tetrahedral intermediate reforms C=O and expels a leaving group. For comparable unactivated derivatives, a useful order is acid chloride > anhydride > ester > amide. Leaving-group basicity and resonance donation both matter.

## acyl-mechanism:cue | Memory Cues

Addition–elimination replaces the acyl substituent while retaining a carbonyl.

## acyl-mechanism:trap | Common Mistakes / MCAT Traps

Aldehydes and ketones usually undergo addition because H− or an ordinary carbanion would be poor leaving groups. Acid/base catalysis and ring strain can change rates. Conjugation does not universally increase carbonyl electrophilicity.

## acyl-mechanism:app | MCAT Applications

Amide resonance makes peptide bonds relatively resistant to uncatalyzed hydrolysis. Strained beta-lactams are more reactive than ordinary unstrained amides.

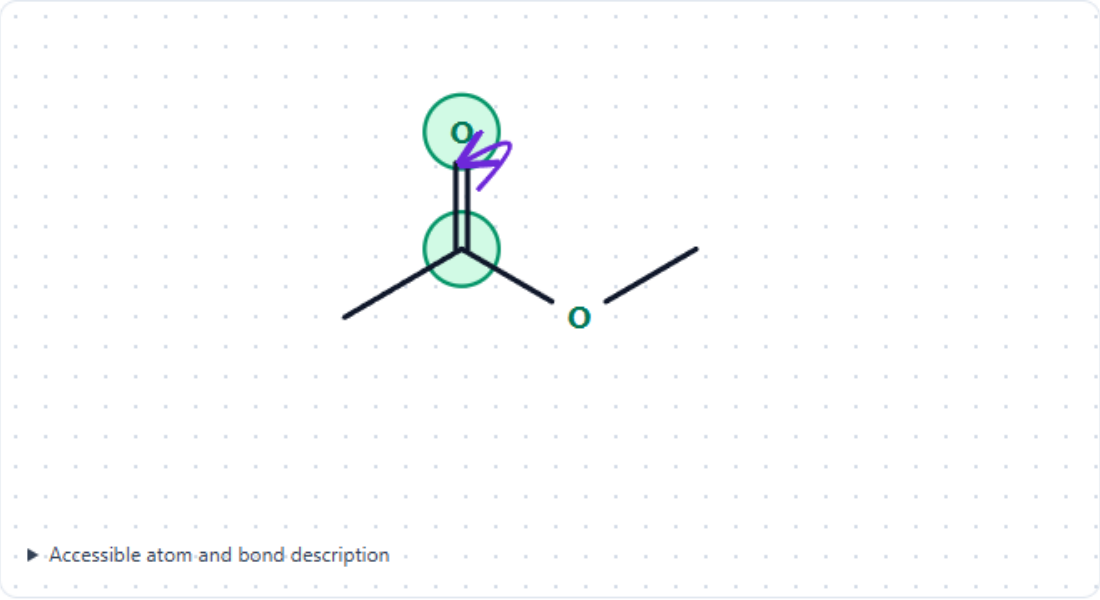
## acyl-mechanism:step-1 | 1. Add to the carbonyl

Hydroxide attacks the ester carbonyl carbon while the pi pair moves to oxygen. The arrow shows that pi-pair movement, not the entire mechanism.

Displayed molecule ID: methylacetate

{"highlights": {"prin": [1, 2], "flow": [{"fromBond": 1, "toAtom": 2}]}, "revealName": false}

Screenshot: acyl-mechanism-step-1



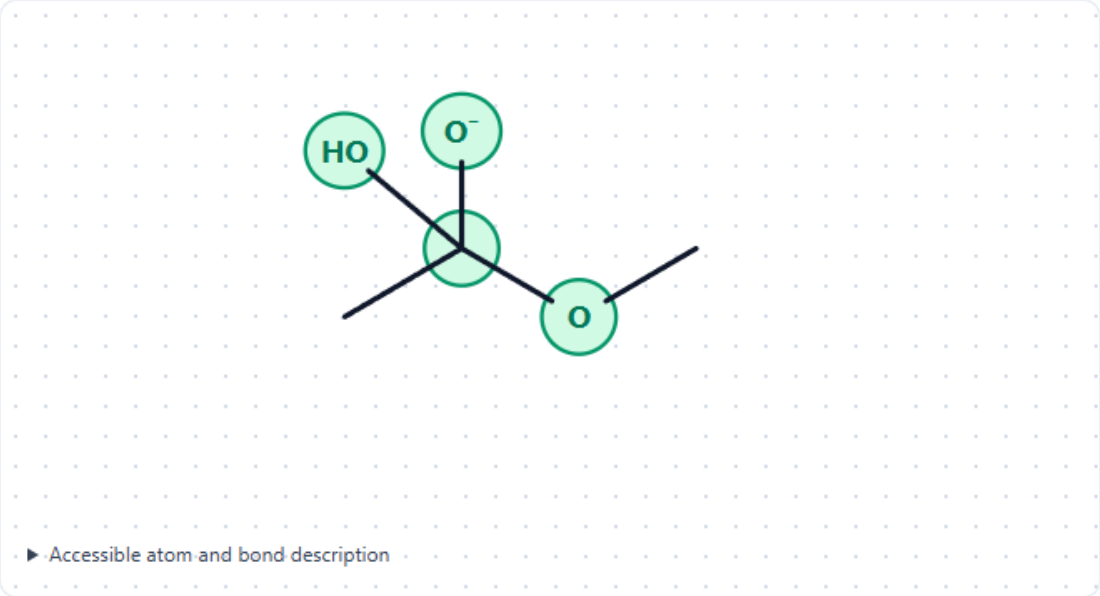

## acyl-mechanism:step-2 | 2. Tetrahedral intermediate

The carbon now bears the incoming OH, the original OCH3, and an O− derived from the carbonyl oxygen. It has four single bonds.

Displayed molecule ID: ester-intermediate

{"highlights": {"prin": [1, 2, 3, 5]}, "revealName": true}

Screenshot: acyl-mechanism-step-2




▶ Accessible atom and bond description

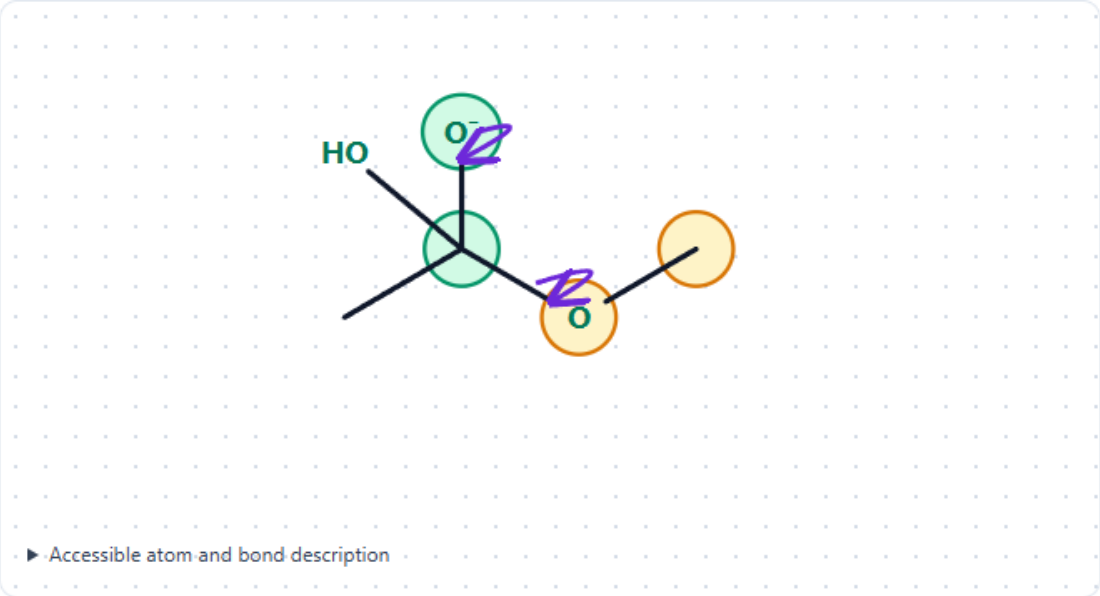
## acyl-mechanism:step-3 | 3. Collapse

An oxygen lone pair reforms C=O, and the C–OCH3 bond pair moves to the leaving oxygen. Subsequent proton transfer yields carboxylate plus methanol under basic conditions.

Displayed molecule ID: ester-intermediate

{"highlights": {"prin": [1, 2], "sub": [3, 4], "flow": [{"fromAtom": 2, "toBond": 1}, {"fromBond": 2, "toAtom": 3}]}, "revealName": false}

Screenshot: acyl-mechanism-step-3




▶ Accessible atom and bond description

# derivative-reactions | Hydrolysis, alcoholysis, and transesterification

Conditions determine the protonation state

## derivative-reactions:rule | Core Rules

Anhydrides react with water to give acids, with alcohols to give ester plus acid, and with amines to give amide plus acid-derived products. Transesterification exchanges one OR group for another under suitable catalysis. Acidic amide hydrolysis gives carboxylic acid and protonated amine; basic hydrolysis gives carboxylate and amine.

## derivative-reactions:cue | Memory Cues

Follow the acyl fragment and the departing group separately.

## derivative-reactions:trap | Common Mistakes / MCAT Traps

In basic ester hydrolysis, carboxylate formation helps drive the net reaction. Do not draw a free carboxylic acid as the major product until acid workup. Amide hydrolysis usually needs more forcing conditions than ester hydrolysis.

## derivative-reactions:app | MCAT Applications

Proteases catalyze peptide hydrolysis under mild biological conditions by activating nucleophiles and stabilizing intermediates; the uncatalyzed reaction is much slower.

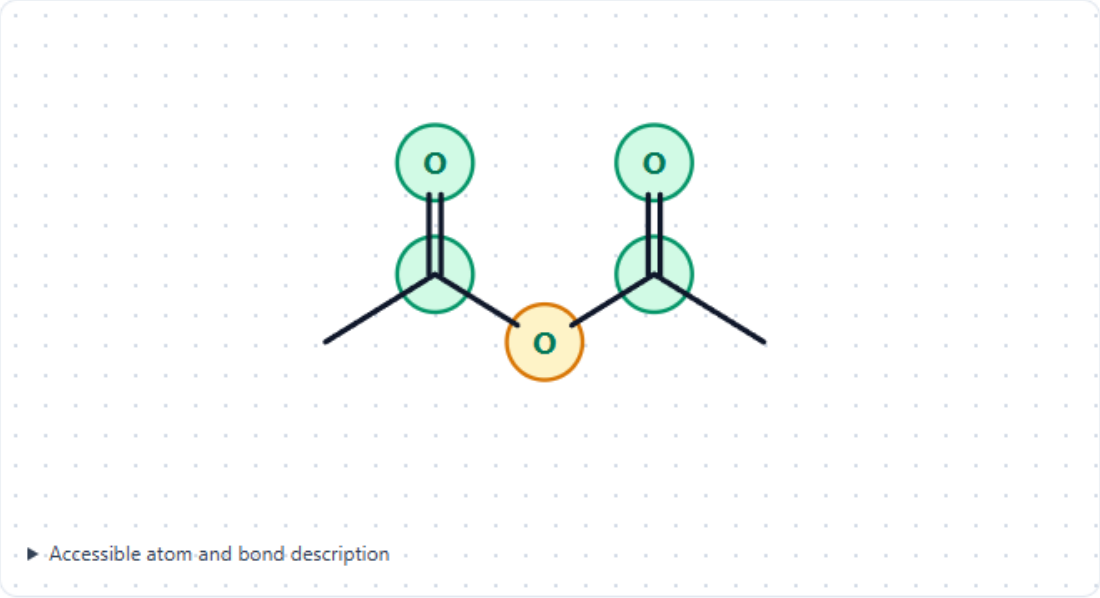
## derivative-reactions:step-1 | 1. Anhydride cleavage

Water attack at either equivalent acyl carbon of ethanoic anhydride leads overall to two ethanoic acid molecules.

Displayed molecule ID: anhyd

{"highlights": {"prin": [1, 3, 5, 6], "sub": [2]}, "revealName": true}

Screenshot: derivative-reactions-step-1



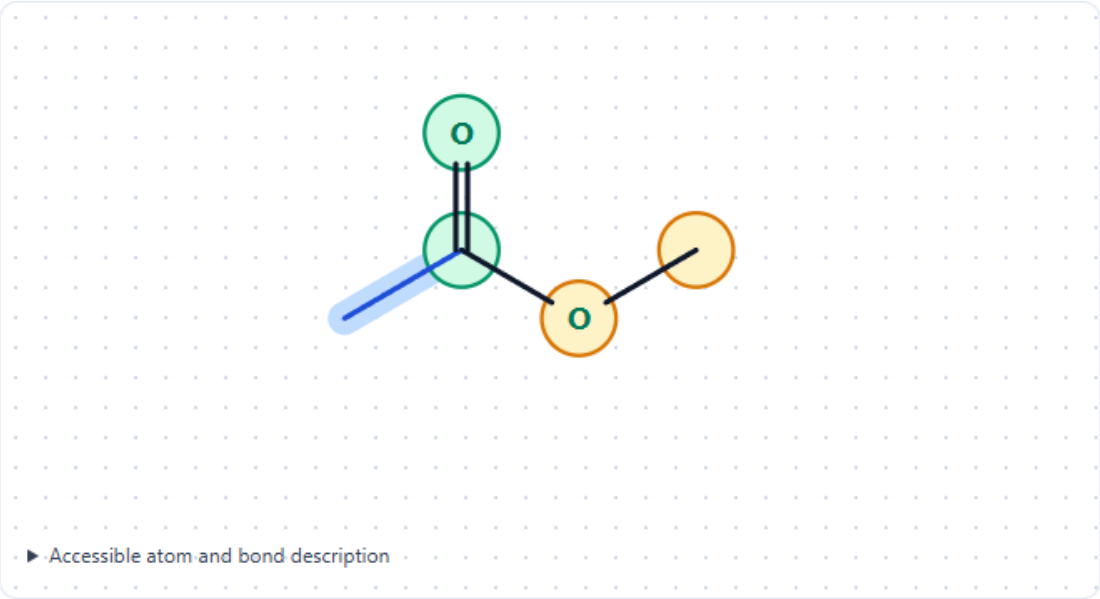

## derivative-reactions:step-2 | 2. Exchange an ester group

In transesterification, the O-alkyl group can be replaced by the incoming alcohol-derived group while the acyl carbon skeleton remains.

Displayed molecule ID: methylacetate

{"highlights": {"path": [0, 1], "prin": [1, 2], "sub": [3, 4]}, "revealName": true}

Screenshot: derivative-reactions-step-2




▶ Accessible atom and bond description

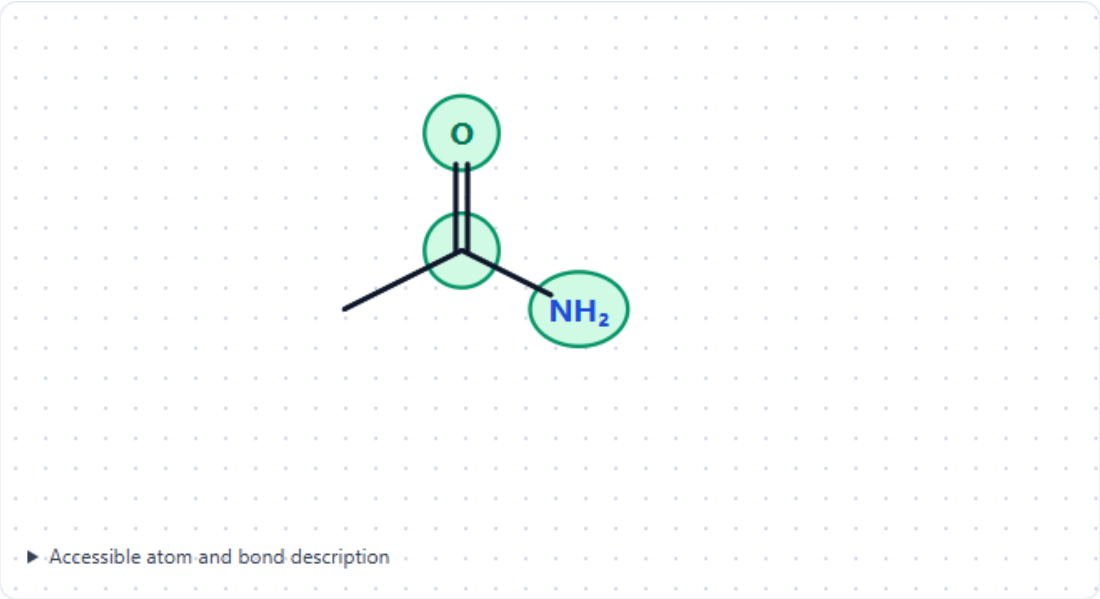
## derivative-reactions:step-3 | 3. Hydrolyze an amide

An amide's N donation lowers electrophilicity and its leaving group is poor unless conditions enable departure. Strong acid/base with heating or a catalyst can accomplish hydrolysis.

Displayed molecule ID: acetamide

{"highlights": {"prin": [1, 2, 3]}, "revealName": true}

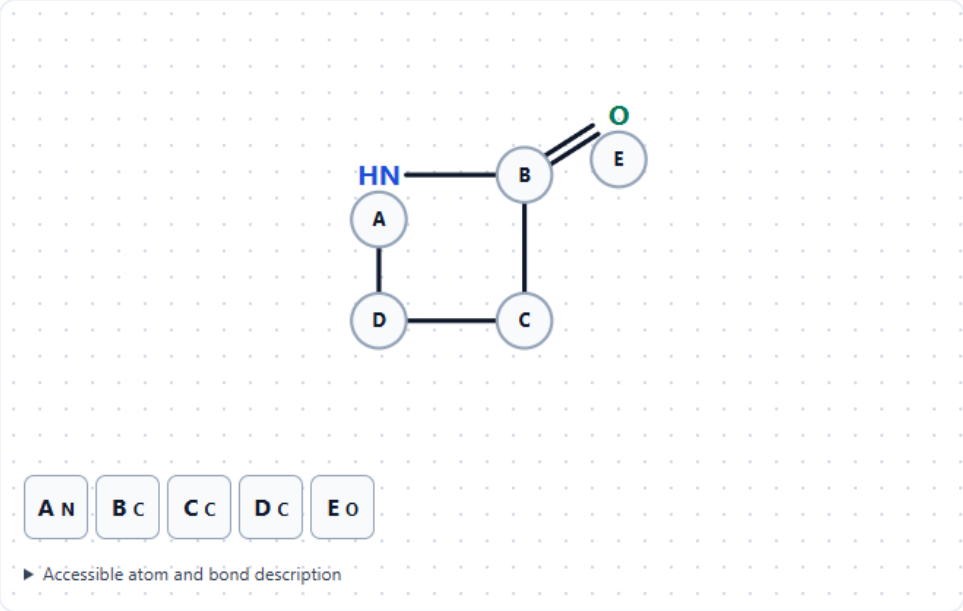
Screenshot: derivative-reactions-step-3




▶ Accessible atom and bond description

# cyclic-derivative | Identify a strained derivative

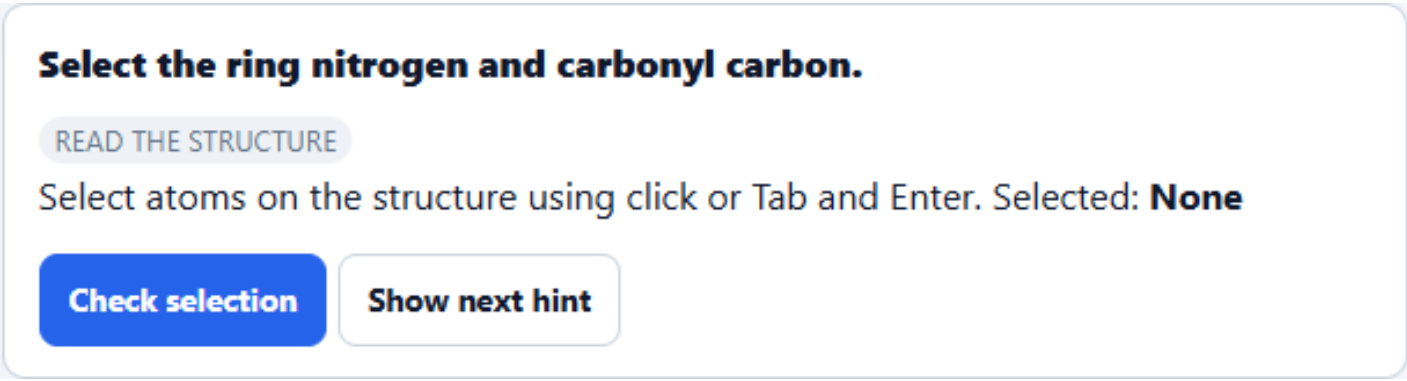
Screenshot: cyclic-derivative-stage-1-molecule




## cyclic-derivative:stage-1 | Select the ring nitrogen and carbonyl carbon.

Answer type: atoms

Screenshot: cyclic-derivative-stage-1-question



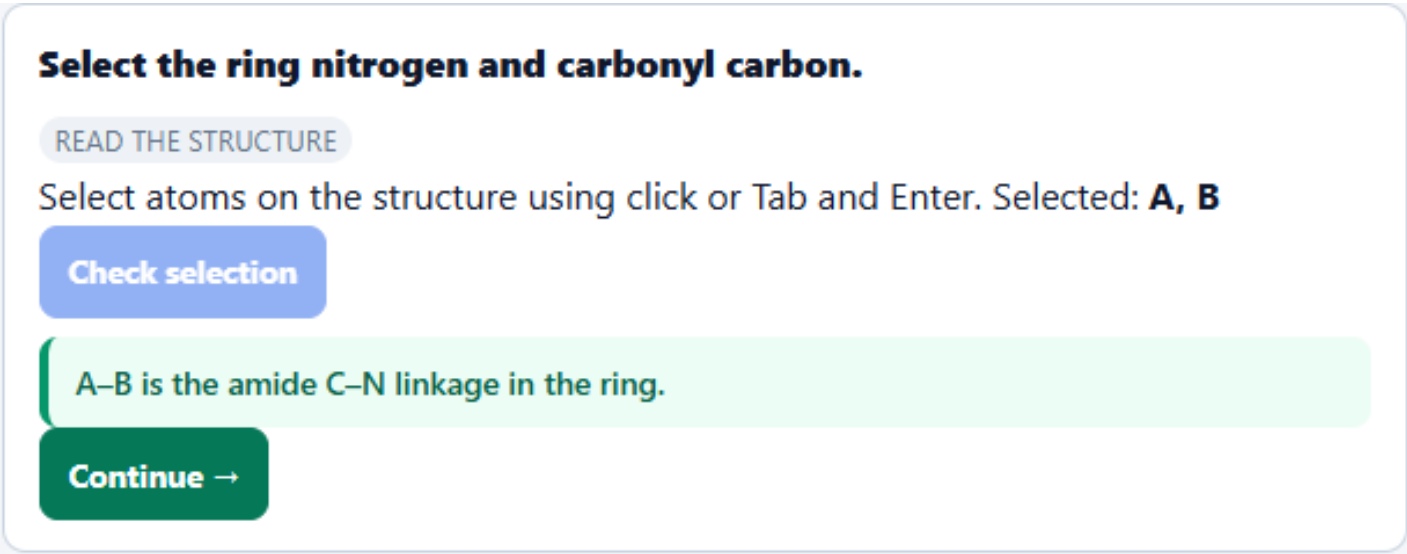

{"authorAnswer": [0, 1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [0, 1]}}

Hint 1: Find N bonded directly to C=O carbon.

Hint 2: Select both endpoints of that bond.

Explanation: A–B is the amide C–N linkage in the ring.

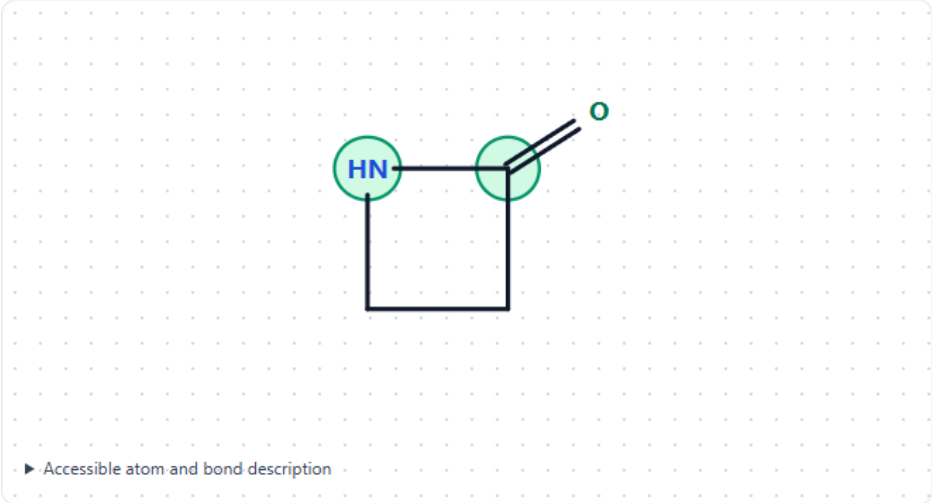
Screenshot: cyclic-derivative-stage-1-feedback



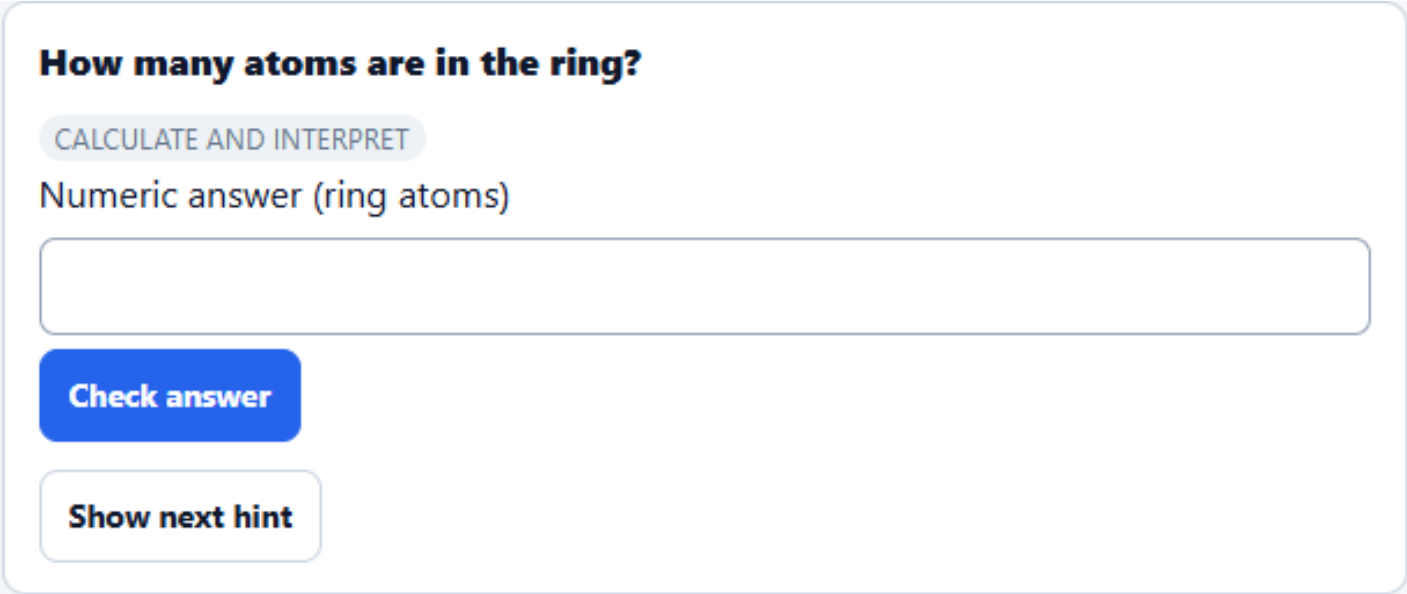

## cyclic-derivative:stage-2 | How many atoms are in the ring?

Answer type: number | unit: ring atoms

Screenshot: cyclic-derivative-stage-2-molecule



Screenshot: cyclic-derivative-stage-2-question



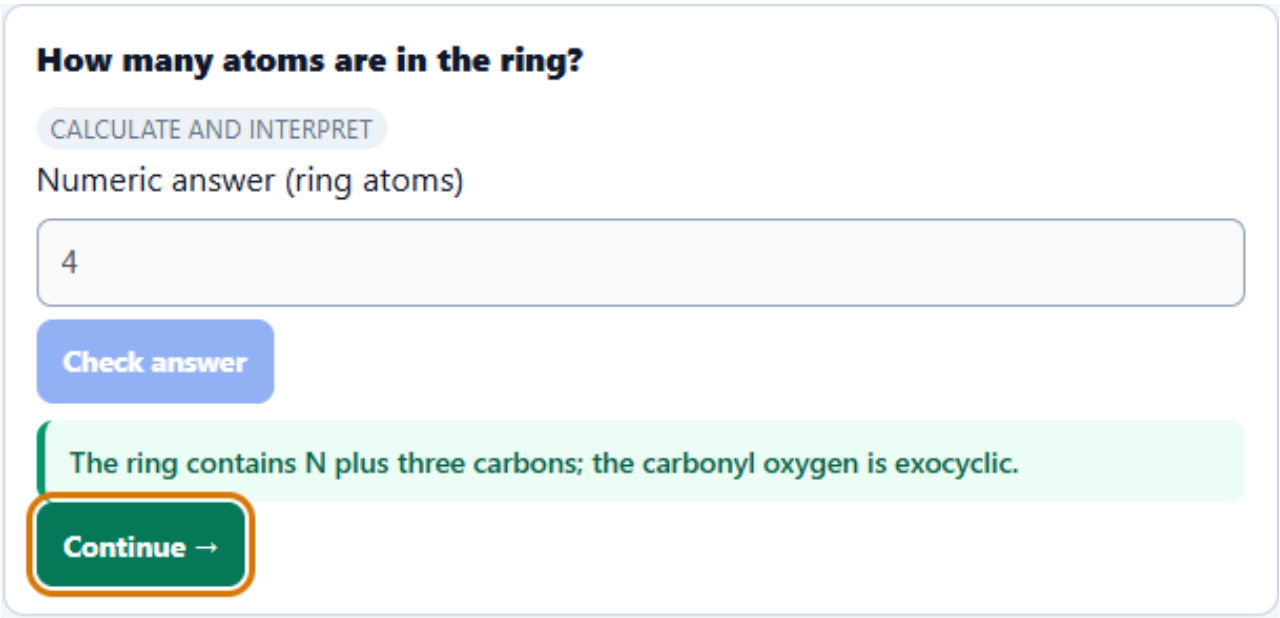

{"authorAnswer": 4, "tolerance": 0, "highlightsAfterCorrectAnswer": {}}

Hint 1: Follow the closed path.

Hint 2: Do not count the oxygen outside the ring.

Explanation: The ring contains N plus three carbons; the carbonyl oxygen is exocyclic.

Screenshot: cyclic-derivative-stage-2-feedback




## cyclic-derivative:stage-3 | The functional-group family is...

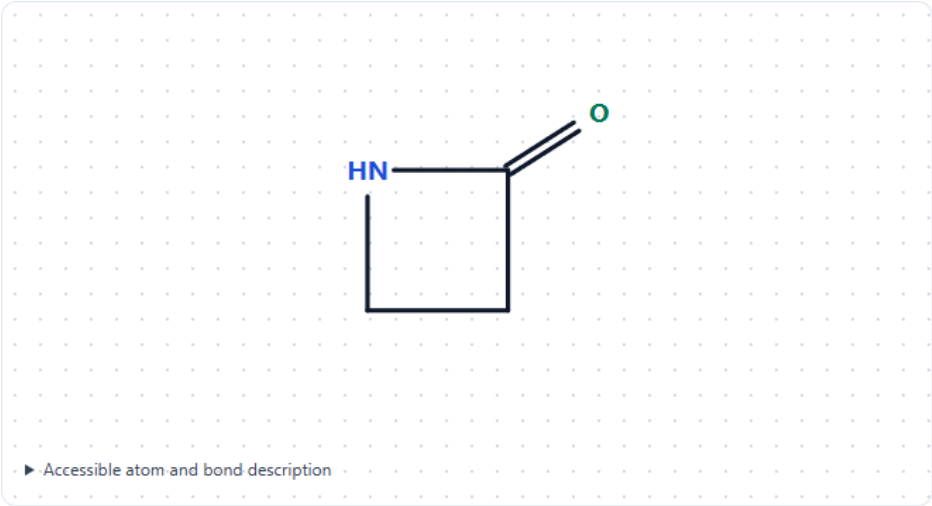
Answer type: choice

Option 0 (A): Lactam

Option 1 (B): Lactone

Option 2 (C): Ether

Screenshot: cyclic-derivative-stage-3-molecule




Screenshot: cyclic-derivative-stage-3-question



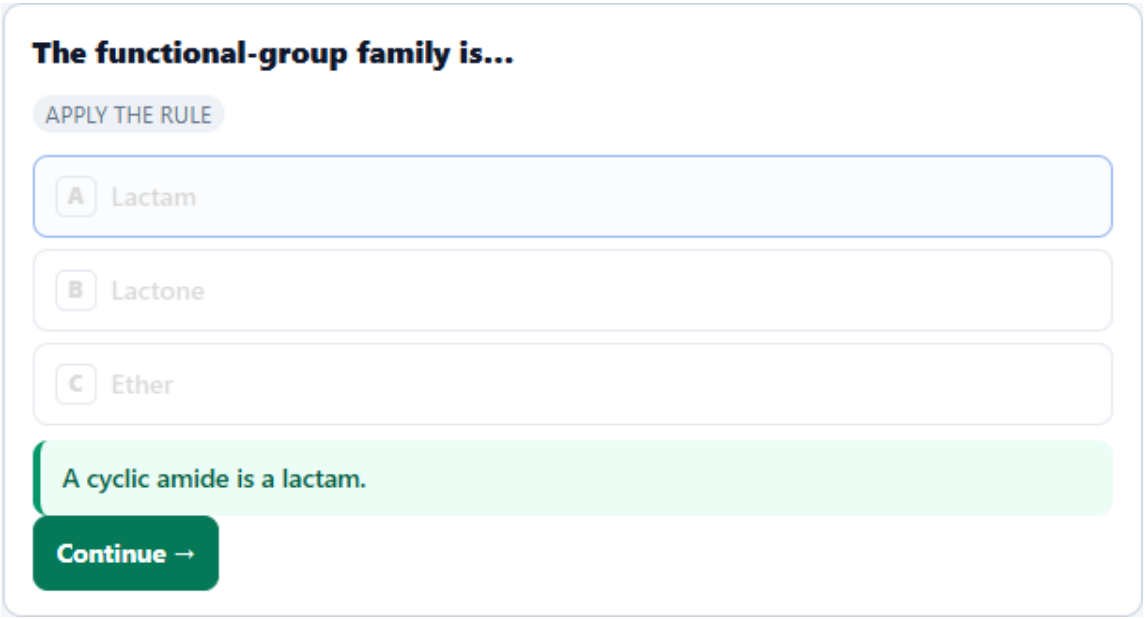

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Use the heteroatom attached to the acyl carbon.

Hint 2: N indicates an amide.

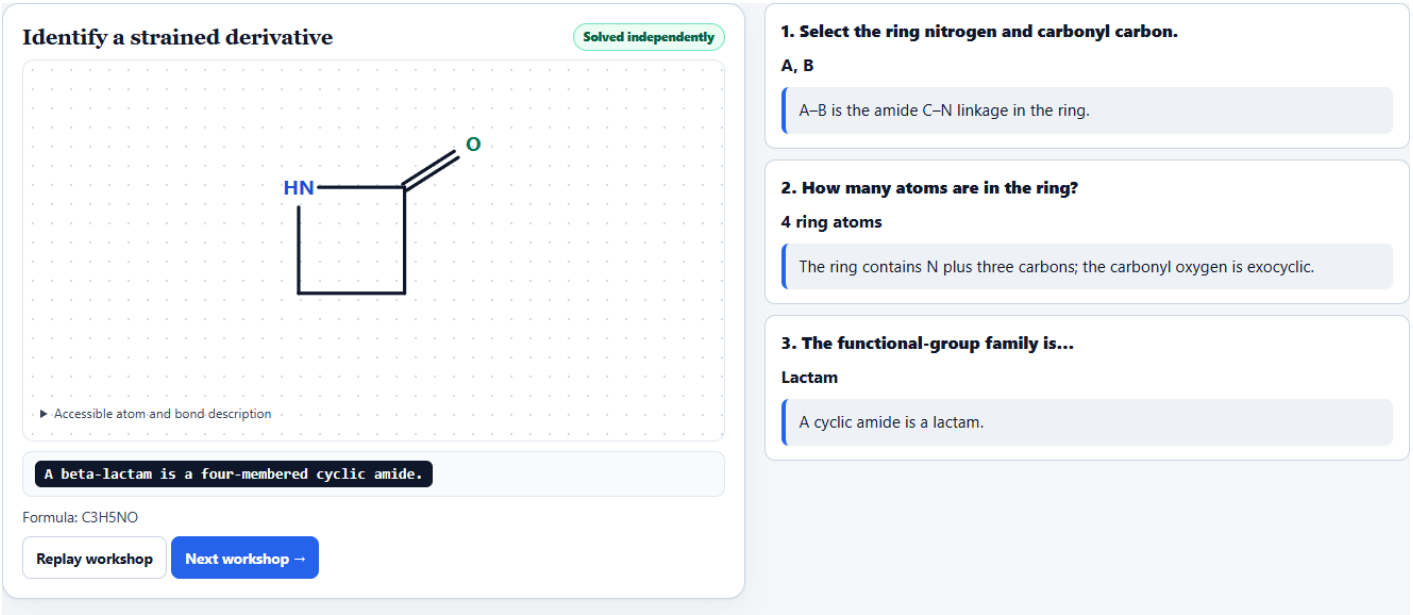
Explanation: A cyclic amide is a lactam.

Screenshot: cyclic-derivative-stage-3-feedback



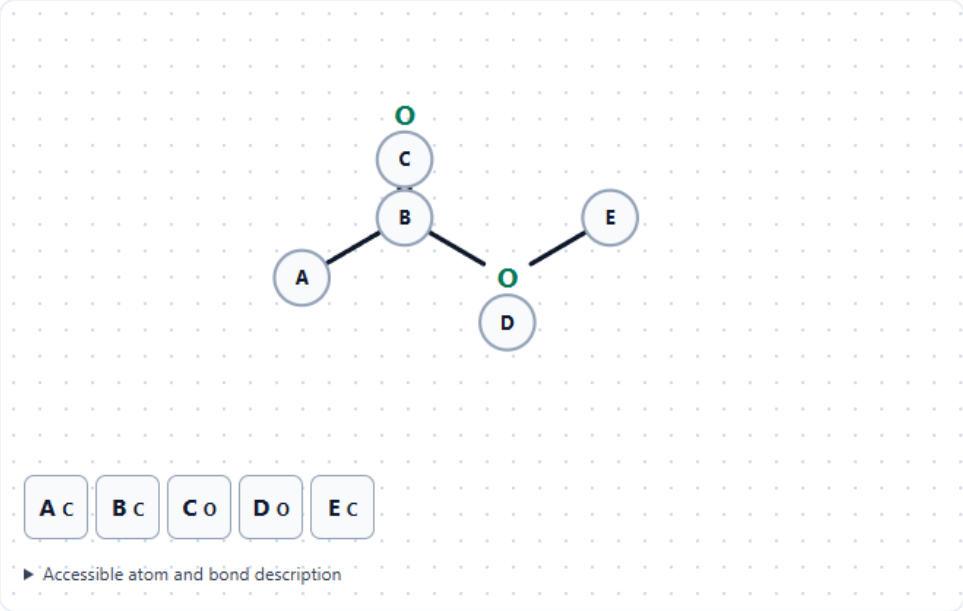

Completion summary: A beta-lactam is a four-membered cyclic amide.

Screenshot: cyclic-derivative-complete



# acyl-order | Follow addition–elimination

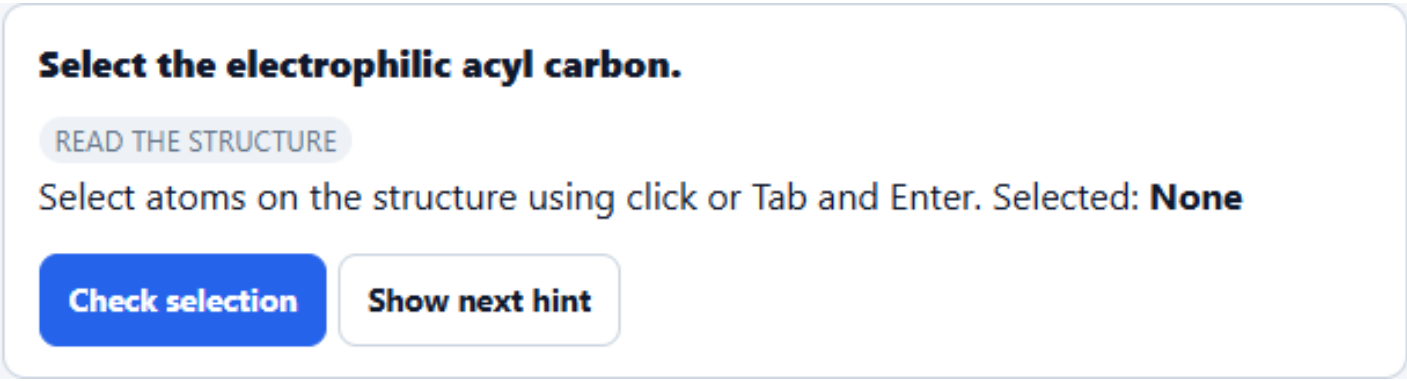
Screenshot: acyl-order-stage-1-molecule




## acyl-order:stage-1 | Select the electrophilic acyl carbon.

Answer type: atoms

Screenshot: acyl-order-stage-1-question



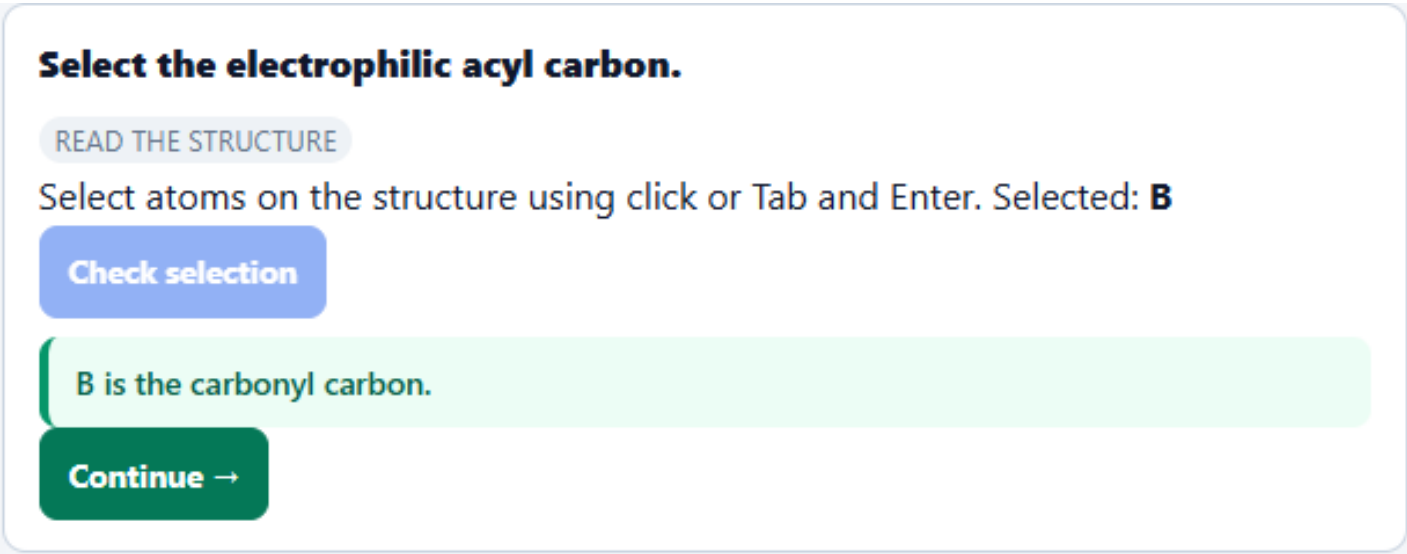

{"authorAnswer": [1], "tolerance": null, "highlightsAfterCorrectAnswer": {"prin": [1, 2]}}

Hint 1: Find C=O.

Hint 2: Choose carbon, not oxygen.

Explanation: B is the carbonyl carbon.

Screenshot: acyl-order-stage-1-feedback




## acyl-order:stage-2 | After addition, what restores the carbonyl?

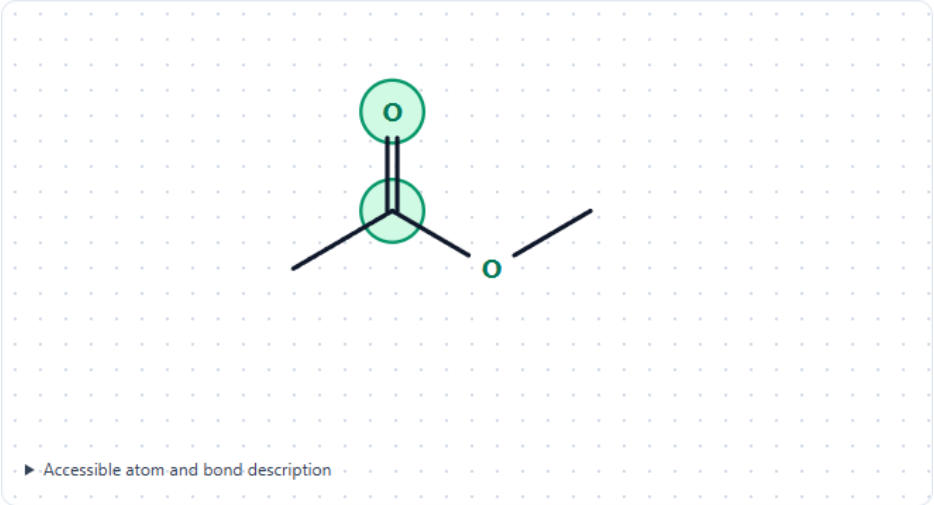
Answer type: choice

Option 0 (A): Collapse of a tetrahedral intermediate with leaving-group departure

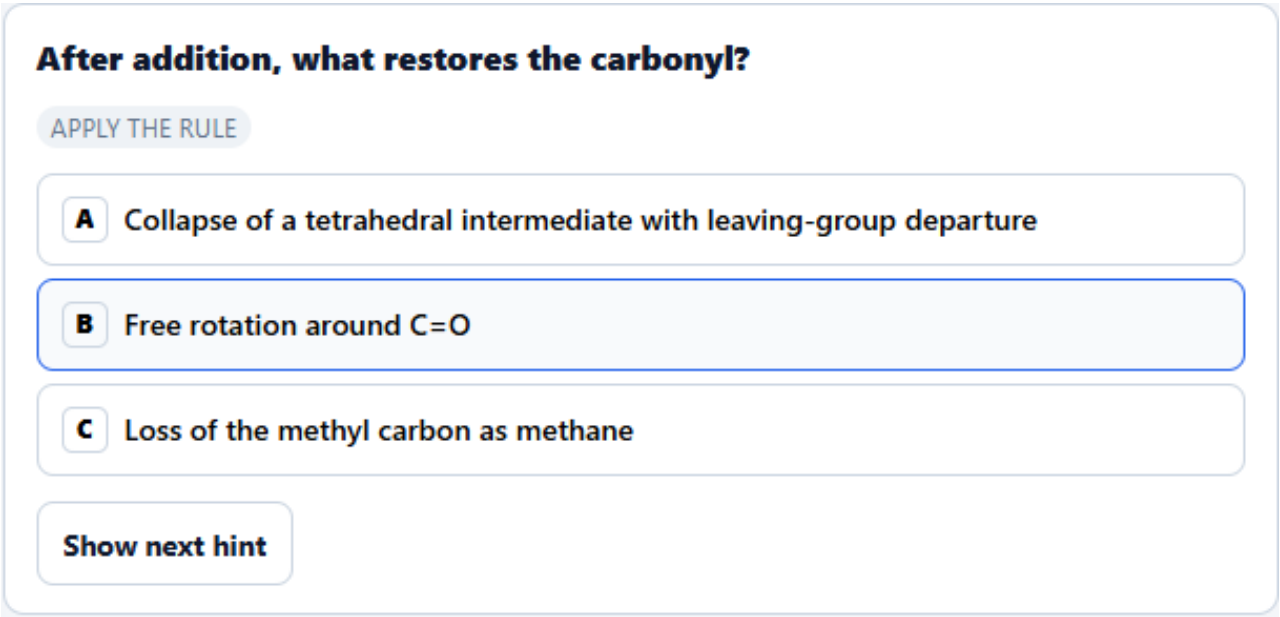
Option 1 (B): Free rotation around C=O

Option 2 (C): Loss of the methyl carbon as methane

Screenshot: acyl-order-stage-2-molecule



Screenshot: acyl-order-stage-2-question



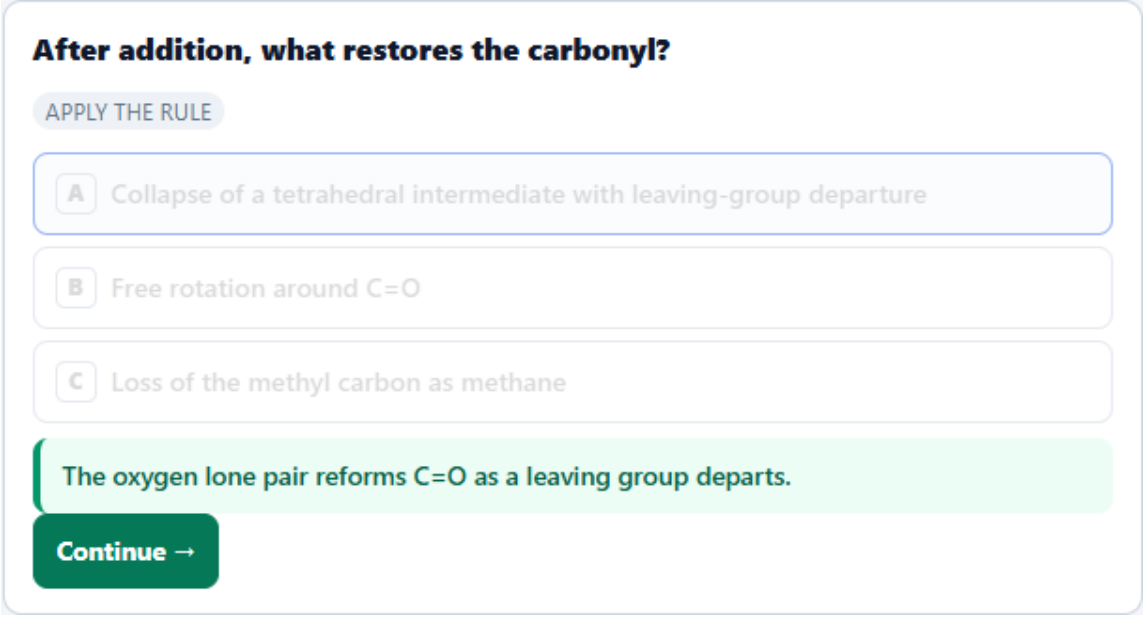

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: The acyl carbon must return to three sigma directions.

Hint 2: Reform the pi bond and expel the heteroatom leaving group.

Explanation: The oxygen lone pair reforms C=O as a leaving group departs.

Screenshot: acyl-order-stage-2-feedback



## acyl-order:stage-3 | Among comparable ordinary anhydrides, esters, and amides, which is least reactive toward acyl substitution?

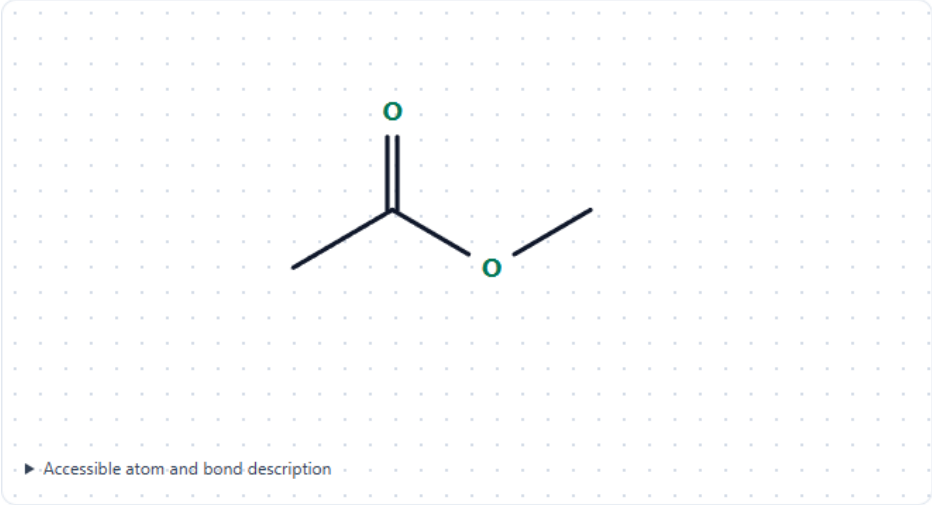
Answer type: choice

Option 0 (A): Anhydride

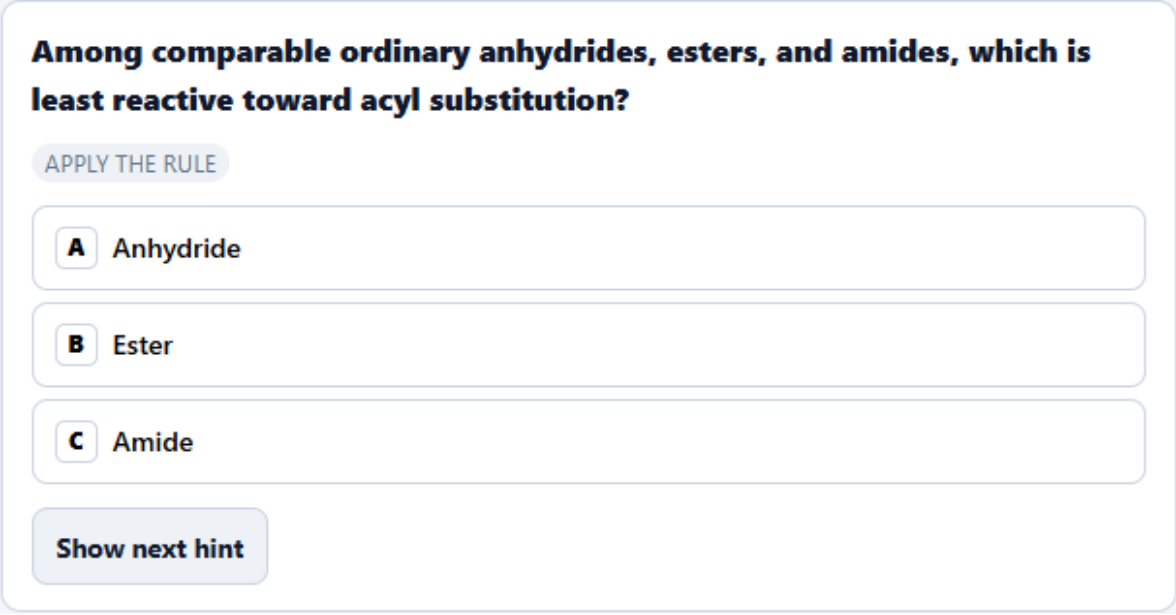
Option 1 (B): Ester

Option 2 (C): Amide

Screenshot: acyl-order-stage-3-molecule




Screenshot: acyl-order-stage-3-question



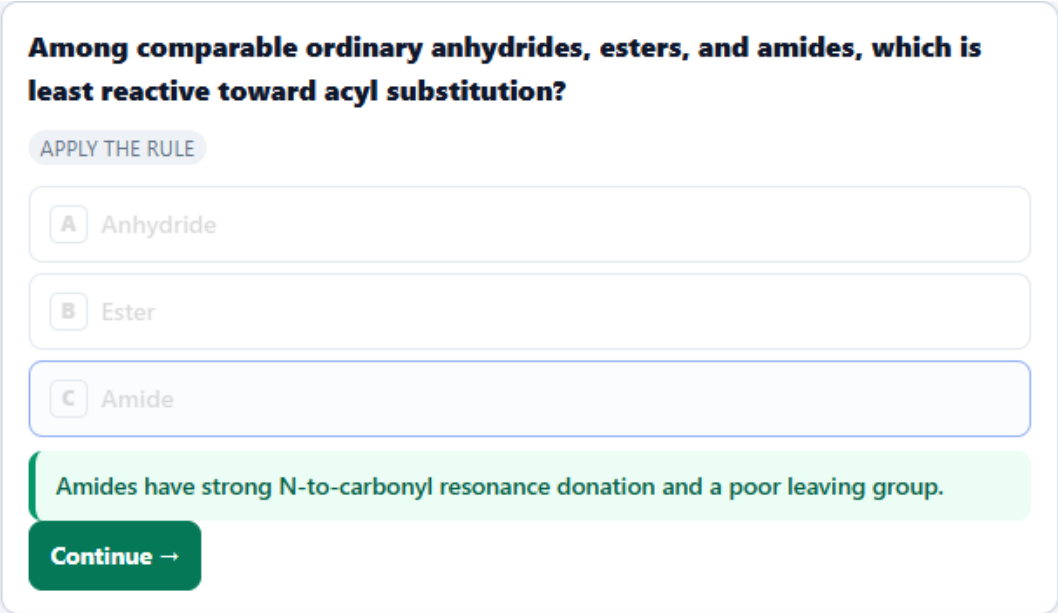

{"authorAnswer": 2, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Consider lone-pair donation and leaving-group basicity.

Hint 2: Nitrogen strongly stabilizes the amide system.

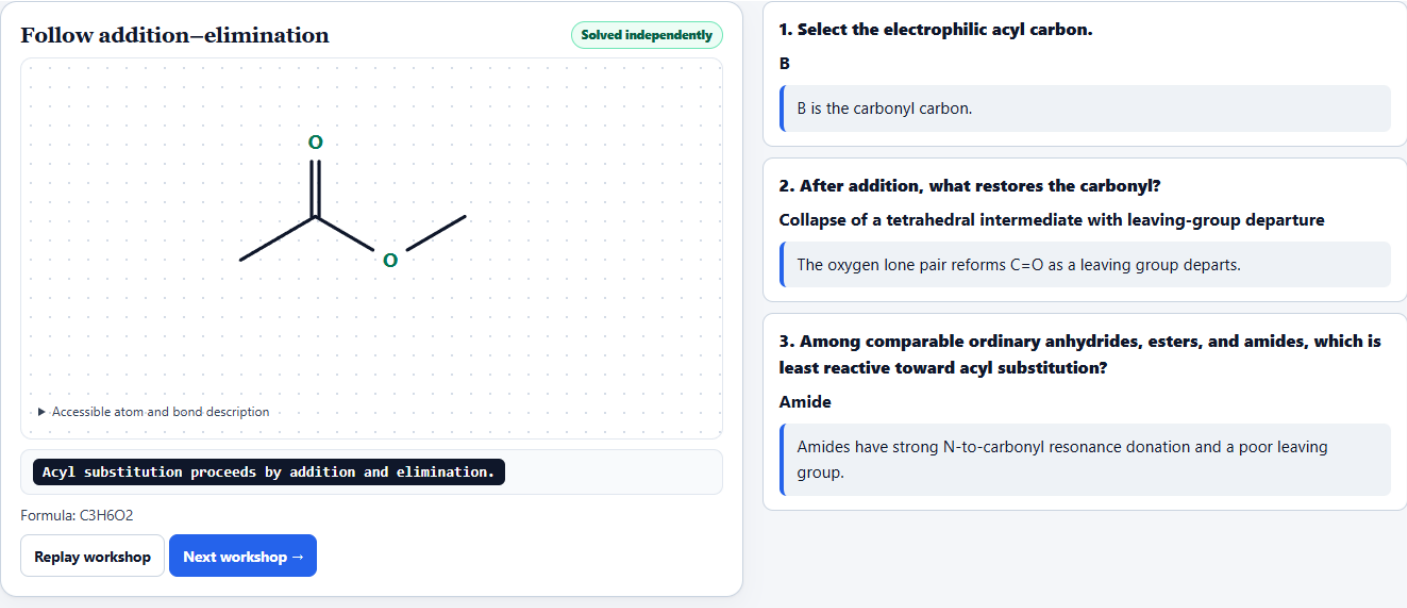
Explanation: Amides have strong N-to-carbonyl resonance donation and a poor leaving group.

Screenshot: acyl-order-stage-3-feedback



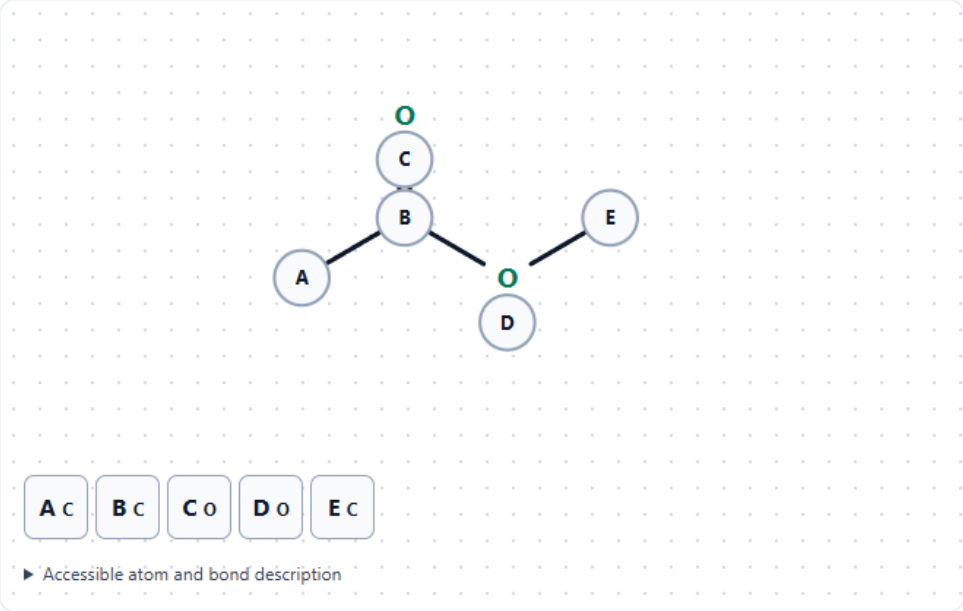

Completion summary: Acyl substitution proceeds by addition and elimination.

Screenshot: acyl-order-complete



# hydrolysis-products | Track hydrolysis and workup

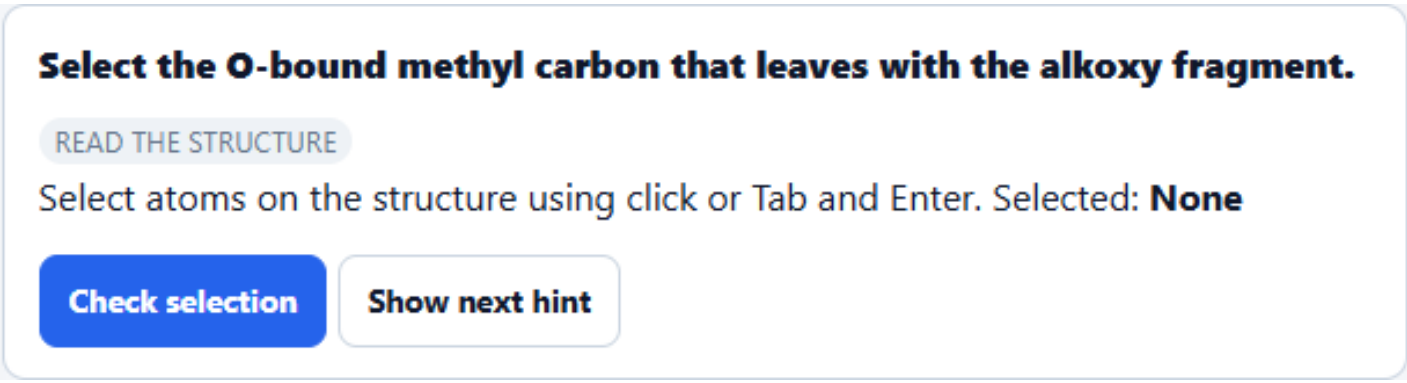
Screenshot: hydrolysis-products-stage-1-molecule




## hydrolysis-products:stage-1 | Select the O-bound methyl carbon that leaves with the alkoxy fragment.

Answer type: atoms

Screenshot: hydrolysis-products-stage-1-question



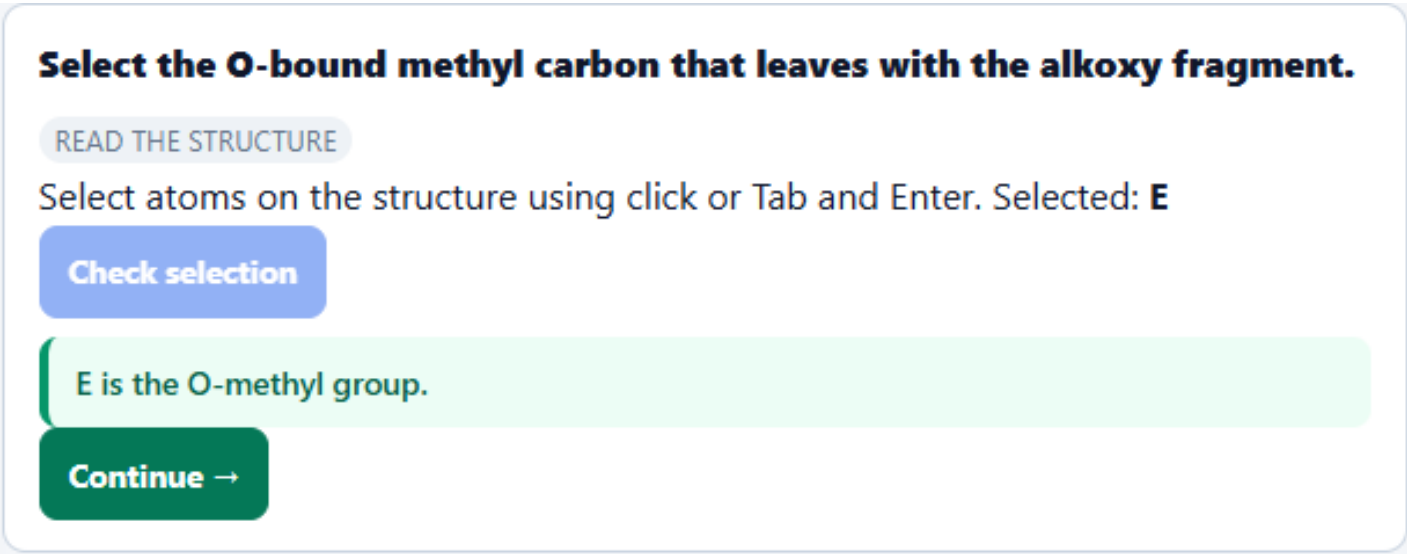

{"authorAnswer": [4], "tolerance": null, "highlightsAfterCorrectAnswer": {"sub": [3, 4]}}

Hint 1: Distinguish the acyl methyl from the O-methyl.

Hint 2: Follow the single-bonded oxygen away from the carbonyl.

Explanation: E is the O-methyl group.

Screenshot: hydrolysis-products-stage-1-feedback




## hydrolysis-products:stage-2 | Under excess aqueous NaOH, the major acid-derived product is…

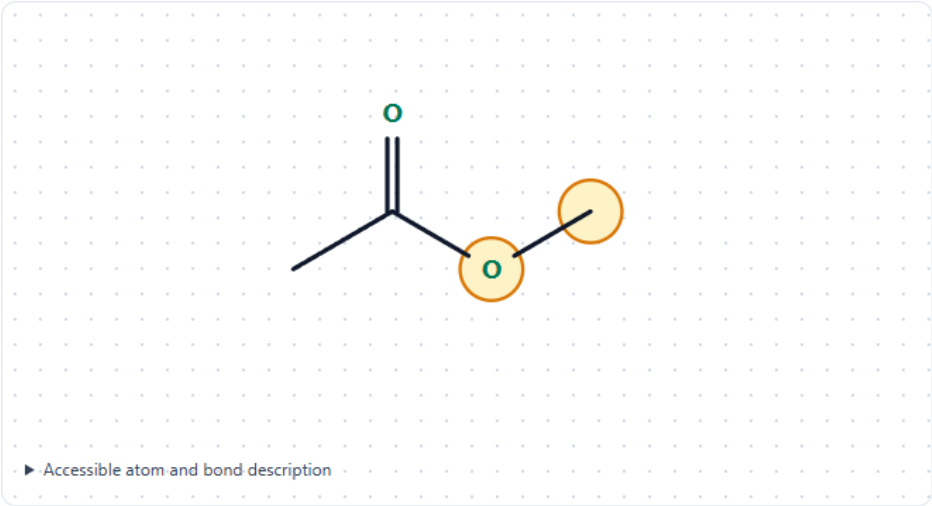
Answer type: choice

Option 0 (A): Acetate

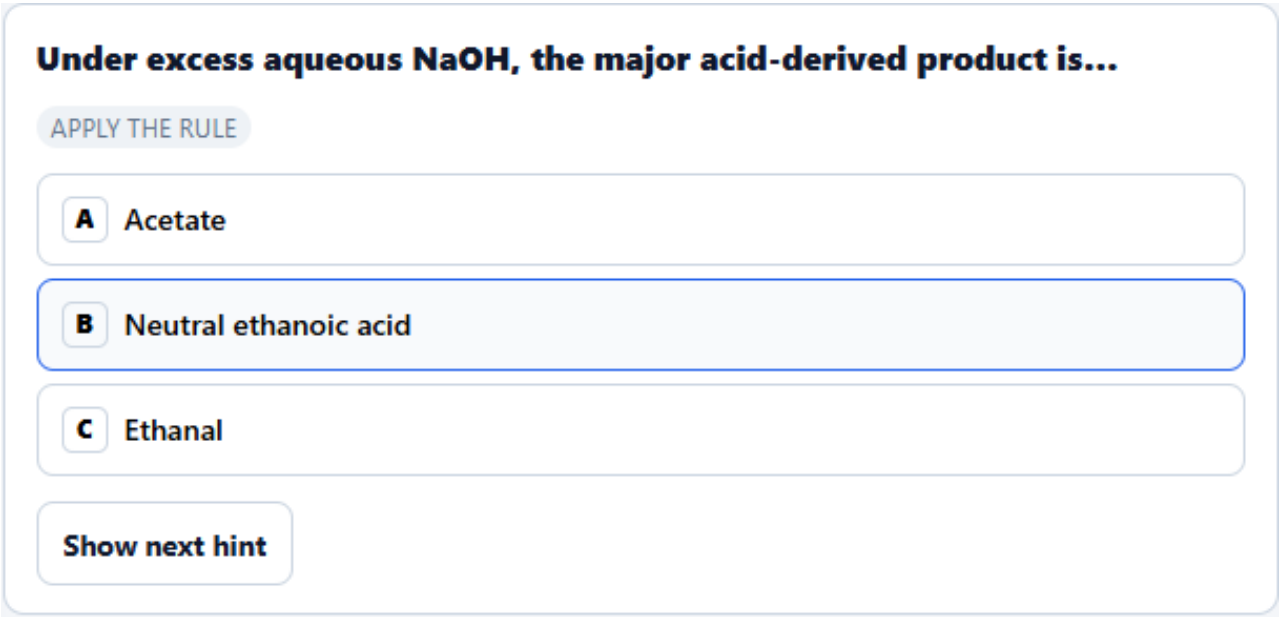
Option 1 (B): Neutral ethanoic acid

Option 2 (C): Ethanal

Screenshot: hydrolysis-products-stage-2-molecule



Screenshot: hydrolysis-products-stage-2-question



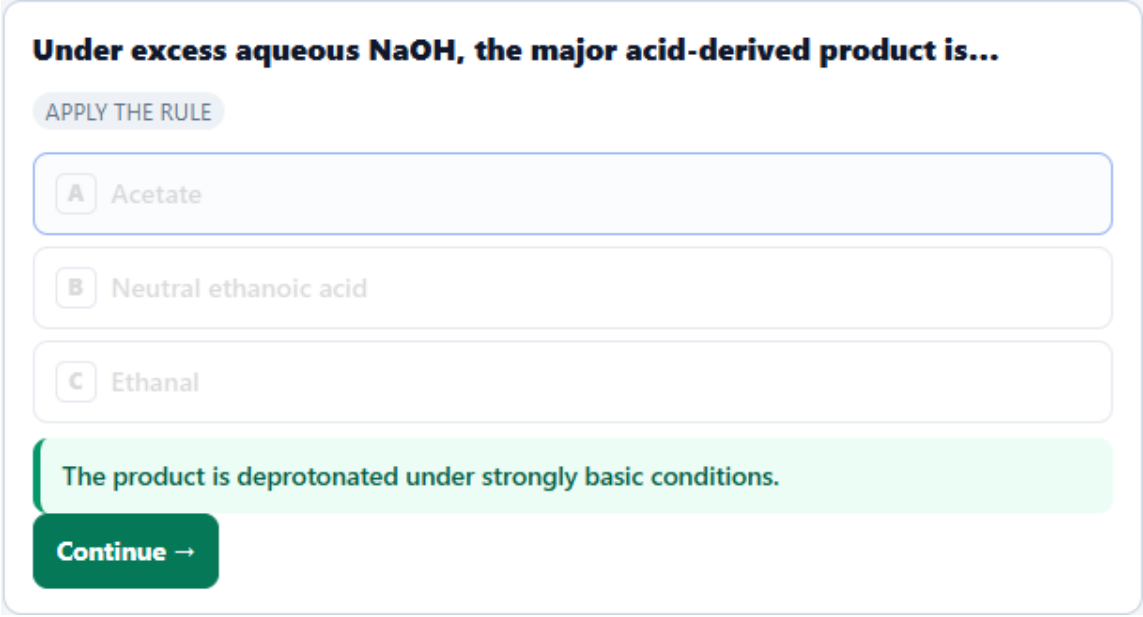

{"authorAnswer": 0, "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Account for acid–base chemistry after substitution.

Hint 2: Carboxylic acids have much lower pKa than water.

Explanation: The product is deprotonated under strongly basic conditions.

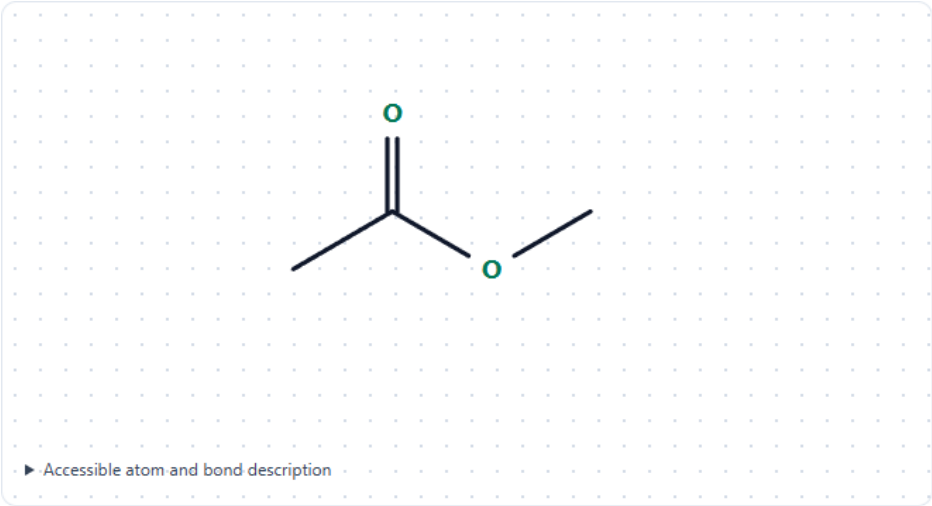
Screenshot: hydrolysis-products-stage-2-feedback




## hydrolysis-products:stage-3 | What alcohol is produced from the O-methyl group?

Answer type: text

Screenshot: hydrolysis-products-stage-3-molecule




Screenshot: hydrolysis-products-stage-3-question

**What alcohol is produced from the O-methyl group?**

STATE YOUR CONCLUSION

Your answer

**Check answer**

**Show next hint**

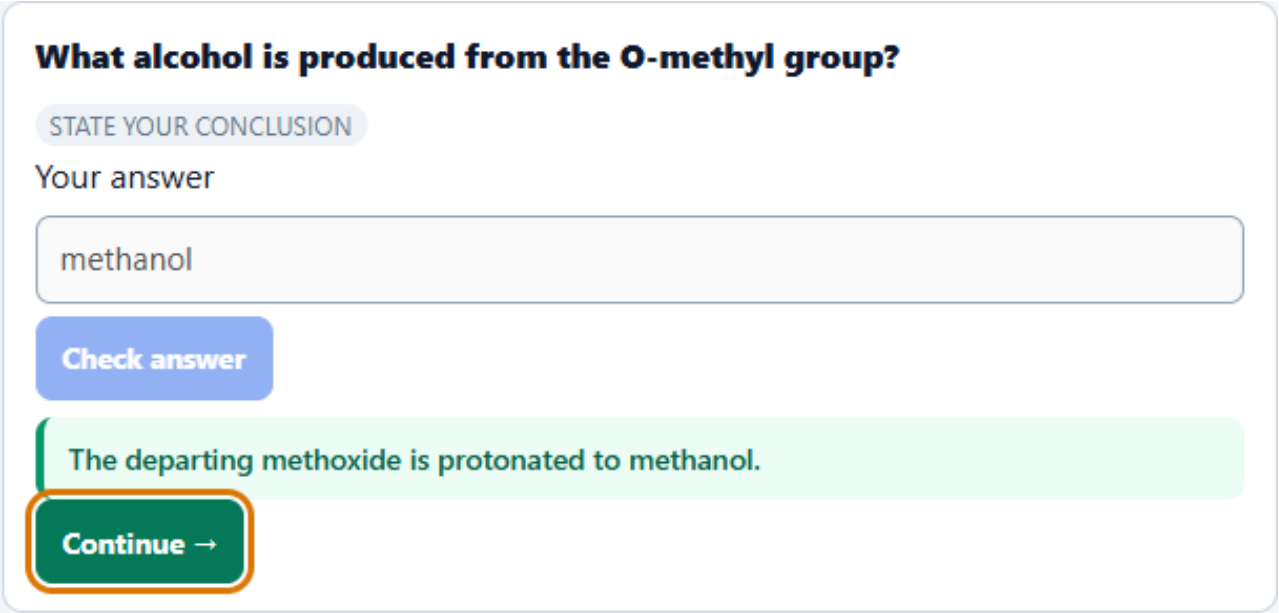
{"authorAnswer": ["methanol", "methyl alcohol"], "tolerance": null, "highlightsAfterCorrectAnswer": {}}

Hint 1: Keep the one-carbon alkoxy fragment.

Hint 2: Replace its bond to acyl carbon with H on oxygen.

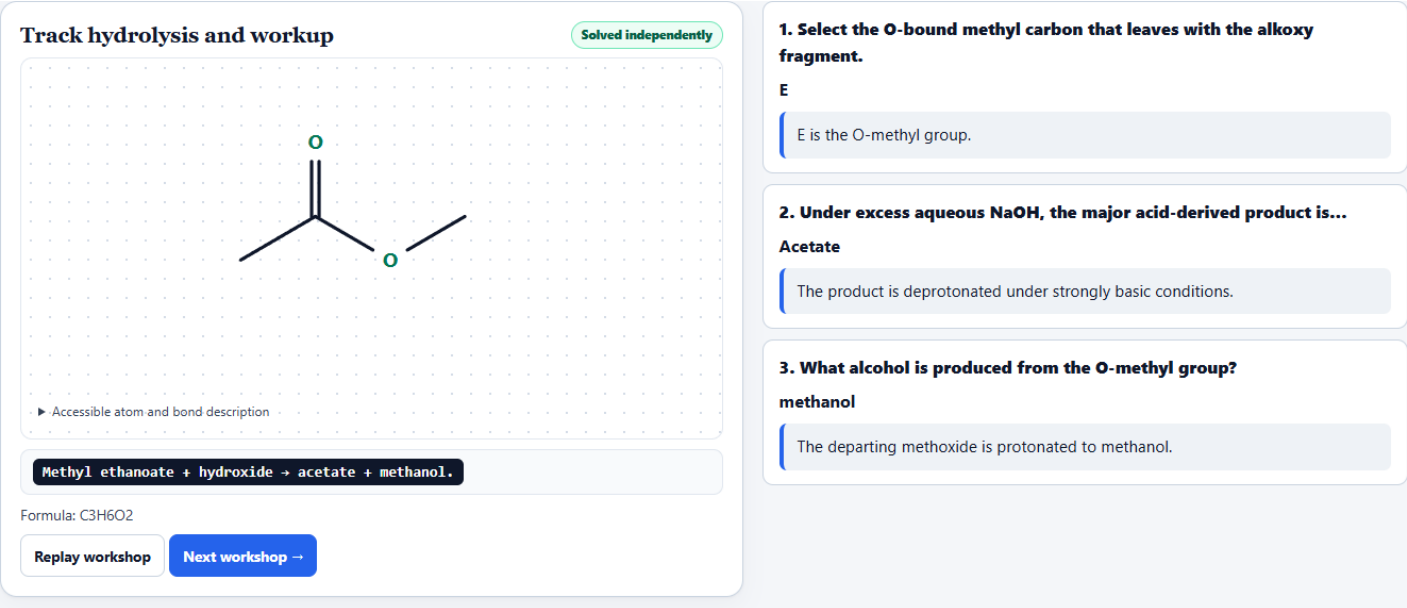
Explanation: The departing methoxide is protonated to methanol.

Screenshot: hydrolysis-products-stage-3-feedback




Completion summary: Methyl ethanoate + hydroxide → acetate + methanol.

Screenshot: hydrolysis-products-complete



# Molecular repository: author claims to verify

## ester

Author claims: ethyl propanoate | formula C5H10O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=156.00, y=52.00; charge=0; explicit H override=none

4 (E): O; x=208.00, y=142.00; charge=0; explicit H override=none

5 (F): C; x=258.00, y=112.00; charge=0; explicit H override=none

6 (G): C; x=308.00, y=142.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [4, 5, 1], [5, 6, 1]]

## betalactam

Author claims: azetidin-2-one (beta-lactam) | formula C3H5NO | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): N; x=100.00, y=60.00; charge=0; explicit H override=none

1 (B): C; x=185.00, y=60.00; charge=0; explicit H override=none

2 (C): C; x=185.00, y=145.00; charge=0; explicit H override=none

3 (D): C; x=100.00, y=145.00; charge=0; explicit H override=none

4 (E): O; x=240.00, y=25.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 0, 1], [1, 4, 2]]

## lactone

Author claims: oxolan-2-one (gamma-butyrolactone) | formula C4H6O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): O; x=135.00, y=35.00; charge=0; explicit H override=none

1 (B): C; x=205.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=180.00, y=165.00; charge=0; explicit H override=none

3 (D): C; x=90.00, y=165.00; charge=0; explicit H override=none

4 (E): C; x=65.00, y=85.00; charge=0; explicit H override=none

5 (F): O; x=265.00, y=55.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [4, 0, 1], [1, 5, 2]]

## methylacetate

Author claims: methyl ethanoate | formula C3H6O2 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

3 (D): O; x=175.00, y=120.00; charge=0; explicit H override=none

4 (E): C; x=235.00, y=85.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1], [3, 4, 1]]

## ester-intermediate

Author claims: tetrahedral intermediate of methyl ethanoate hydroxide addition | formula C3H7O3 | net charge -1

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=120.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=-1; explicit H override=none

3 (D): O; x=175.00, y=120.00; charge=0; explicit H override=none

4 (E): C; x=235.00, y=85.00; charge=0; explicit H override=none

5 (F): O; x=55.00, y=35.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [3, 4, 1], [1, 5, 1]]

## anhyd

Author claims: ethanoic anhydride | formula C4H6O3 | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=45.00, y=130.00; charge=0; explicit H override=none

1 (B): C; x=100.00, y=96.00; charge=0; explicit H override=none

2 (C): O; x=155.00, y=130.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=96.00; charge=0; explicit H override=none

4 (E): C; x=265.00, y=130.00; charge=0; explicit H override=none

5 (F): O; x=100.00, y=40.00; charge=0; explicit H override=none

6 (G): O; x=210.00, y=40.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [1, 5, 2], [3, 6, 2]]

## acetamide

Author claims: ethanamide | formula C2H5NO | net charge +0

Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=55.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=115.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=115.00, y=25.00; charge=0; explicit H override=none

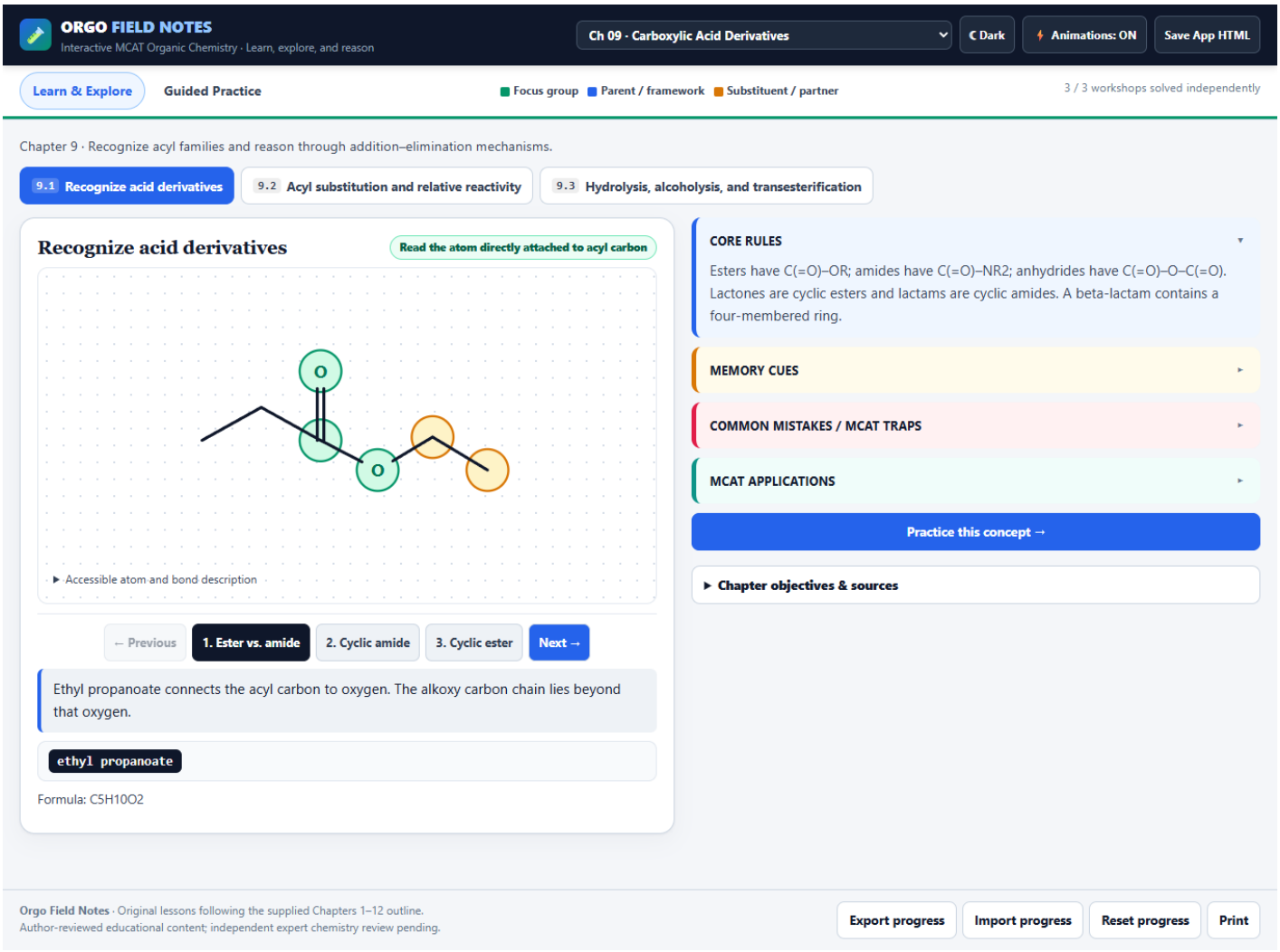
3 (D): N; x=175.00, y=115.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 2], [1, 3, 1]]

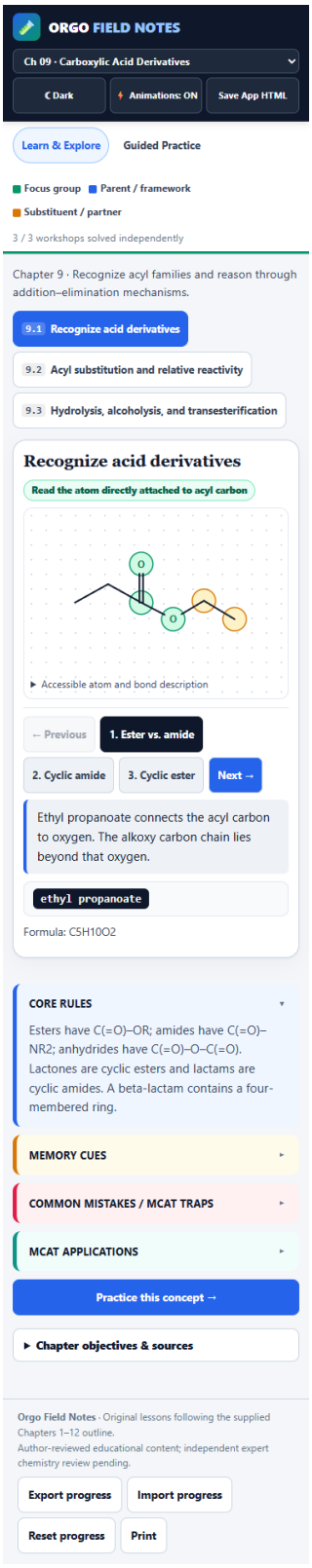
# Layout samples

These overview images are scaled for page fit. Detailed structure and stage images appear earlier. Samples cover the first concept on desktop and mobile in both themes, not every responsive state.

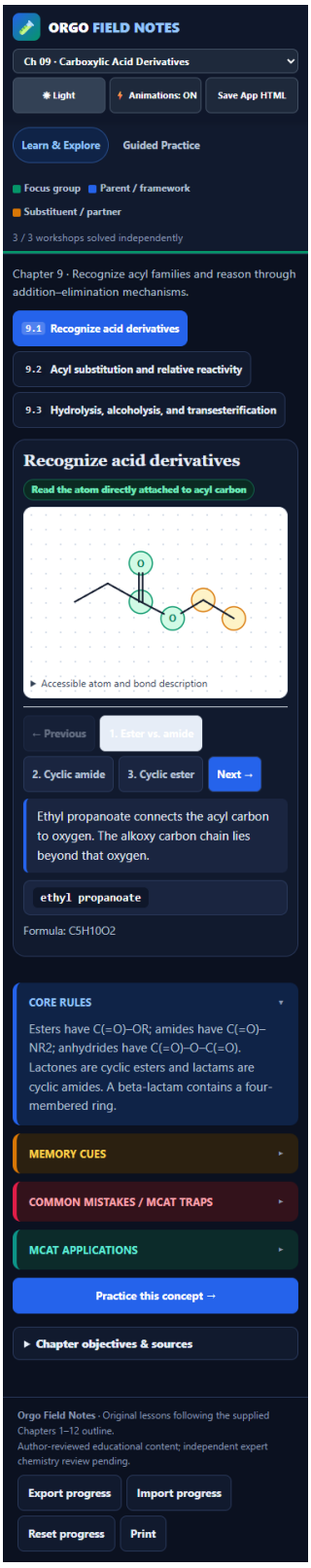
Screenshot: layout-desktop-light



Screenshot: layout-mobile-light



Screenshot: layout-mobile-dark




# Execution evidence

42 screenshots and 9 recorded interactions. Incorrect answers rejected and encoded keys accepted for captured guided stages. This checks implementation against its data; it does not validate chemistry independently. Static screenshots do not verify animation timing, arbitrary clicks, or screen-reader experience.